

The clustering window of the Gaussian covariance beyond the local approximation

From definitions to an implementation in `thecov`

Validation note, branch `thecov2` — October 2026

Summary. The Gaussian covariance of a windowed power spectrum contains, in its clustering terms, the product of the window at the two ends of a galaxy pair, $m(\mathbf{x})m(\mathbf{x} + \mathbf{r})$, weighted by the correlation function $\xi(\mathbf{r})$. `thecov` replaces this by $m^2(\mathbf{x})$ (the *local approximation*). Near window structure smaller than a correlation length — veto holes, footprint edges, completeness patches — this is wrong, and for DESI LRG1 it biases the Gaussian variance low by $\sim 7\%$. The correct clustering window is $W_k(\mathbf{x}) = m(\mathbf{x})(K_k \star m)(\mathbf{x})$, with a kernel K_k fixed by ξ itself, so no free smoothing scale is introduced. The same quantity sets the dilution of the measured power spectrum by the window, $I_k = \int W_k$, which replaces the single number $\int m^2$ in the model normalisation. We give the corrections for a theory and for a measured model power spectrum, the evidence from the DESI DR2 validation, and an implementation.

1 Definitions

Catalogues and weights. Galaxies g at $\mathbf{x}_g$ with total weights w_g (`WEIGHT` $\times$ `WEIGHT_FKP`), randoms r with weights w_r , and $\alpha = \sum_g w_g / \sum_r w_r$. The weighted overdensity field is

$$F(\mathbf{x}) = \sum_g w_g \delta_D(\mathbf{x} - \mathbf{x}_g) - \alpha \sum_r w_r \delta_D(\mathbf{x} - \mathbf{x}_r). \quad (1)$$

Mean weighted density. With weights treated as random variables, independent between objects and of the density field,

$$m(\mathbf{x}) = \left\langle \sum_g w_g \delta_D(\mathbf{x} - \mathbf{x}_g) \right\rangle = \bar{n}(\mathbf{x}) \langle w \rangle(\mathbf{x}), \quad (2)$$

where $\langle w \rangle(\mathbf{x})$ is the local mean weight. Clustering terms involve $\langle w \rangle$; shot-noise terms involve $\langle w^2 \rangle$ through the shot-noise density

$$S(\mathbf{x}) = (1 + \alpha) \bar{n}(\mathbf{x}) \langle w^2 \rangle(\mathbf{x}), \quad (3)$$

normalised so that $\int S = \text{num_shotnoise}$, the realised shot noise the spectra subtract.

Estimator. With the normalisation `norm` of the spectrum files (computed by `jaxpower` on $10h^{-1}\text{Mpc}$ cells),

$$\hat{P}(\mathbf{k}) = \frac{|\tilde{F}(\mathbf{k})|^2 - \int S}{\text{norm}}, \quad (4)$$

averaged over a k -shell and, for multipoles, weighted by Legendre polynomials of the line of sight (we keep the monopole notation; the multipole case follows with the usual $\mathcal{L}_\ell$ factors).

2 The exact Gaussian covariance

For a Gaussian overdensity δ with correlation function ξ and power spectrum P , the two-point function of $\tilde{F}$ has a clustering and a shot-noise part,

$$\langle \tilde{F}(\mathbf{k}) \tilde{F}^*(\mathbf{k}') \rangle = G_P(\mathbf{k}, \mathbf{k}') + G_S(\mathbf{k} - \mathbf{k}'), \quad (5)$$

$$G_P(\mathbf{k}, \mathbf{k}') = \int d^3\mathbf{x} d^3\mathbf{y} m(\mathbf{x})m(\mathbf{y})\xi(\mathbf{x} - \mathbf{y})e^{-i\mathbf{k}\cdot\mathbf{x} + i\mathbf{k}'\cdot\mathbf{y}}, \quad G_S(\mathbf{q}) = \int d^3\mathbf{x} S(\mathbf{x})e^{-i\mathbf{q}\cdot\mathbf{x}}. \quad (6)$$

By Wick's theorem the covariance of the estimator is

$$\text{Cov}[\hat{P}(k), \hat{P}(k')] = \frac{1}{\text{norm}^2} \langle |G_P(\mathbf{k}, \mathbf{k}') + G_S(\mathbf{k} - \mathbf{k}')|^2 + (\mathbf{k}' \rightarrow -\mathbf{k}') \rangle_{k, k' \text{ shells}}, \quad (7)$$

which expands into three terms: PP ($|G_P|^2$), PS ($2 \text{Re } G_P G_S^*$) and SS ($|G_S|^2$). Equation 7 is exact for a Gaussian field; everything below concerns how G_P is evaluated.

3 The local approximation and its failure

Write $\mathbf{y} = \mathbf{x} + \mathbf{r}$ in G_P :

$$G_P(\mathbf{k}, \mathbf{k}') = \int d^3\mathbf{x} e^{-i(\mathbf{k} - \mathbf{k}')\cdot\mathbf{x}} \int d^3\mathbf{r} \xi(\mathbf{r}) e^{i\mathbf{k}'\cdot\mathbf{r}} m(\mathbf{x})m(\mathbf{x} + \mathbf{r}). \quad (8)$$

The $\mathbf{r}$ integral extends over a correlation length (tens of $h^{-1}\text{Mpc}$ for the k of interest). If m is smooth on that scale, $m(\mathbf{x} + \mathbf{r}) \approx m(\mathbf{x})$, the $\mathbf{r}$ integral gives $P(k')$, and

$$G_P(\mathbf{k}, \mathbf{k}') \approx P(k) \tilde{W}(\mathbf{k} - \mathbf{k}'), \quad W(\mathbf{x}) = m^2(\mathbf{x}). \quad (9)$$

This is the local approximation used by `thecov` (and by FFT-mesh implementations of the Gaussian covariance). It fails wherever m changes within a correlation length. DESI footprints contain many such structures: bright-star and other veto masks of arcminute size (below $1h^{-1}\text{Mpc}$), completeness variations on tile scales, and footprint edges. Since the randoms sit only where $m > 0$, `thecov` evaluates m^2 between the holes, as if the partner of every pair were also between the holes.

4 The pair-averaged window

Keep $m(\mathbf{x} + \mathbf{r})$ inside the $\mathbf{r}$ integral of Equation 8 and use only $k \approx k'$ within a window width (the same order of approximation as Equation 9):

$$G_P(\mathbf{k}, \mathbf{k}') \approx P(k) \tilde{W}_k(\mathbf{k} - \mathbf{k}'), \quad W_k(\mathbf{x}) = m(\mathbf{x}) (K_k \star m)(\mathbf{x}), \quad (10)$$

$$K_k(\mathbf{r}) = \frac{\xi(\mathbf{r})e^{i\mathbf{k}\cdot\mathbf{r}}}{P(k)}, \quad \int d^3\mathbf{r} K_k(\mathbf{r}) = 1. \quad (11)$$

The correct clustering window is m times m averaged over the partner's position with the kernel K_k . For an isotropic estimate one uses the angle average,

$$K_k(r) = \frac{\xi(r)j_0(kr)}{P(k)}, \quad \int_0^\infty 4\pi r^2 K_k(r) dr = 1. \quad (12)$$

Two consequences:

1. **No free smoothing scale.** The averaging is set by ξ ; its radial weight $4\pi r^2 \xi(r)j_0(kr)$ extends to $r \sim 10\text{--}30h^{-1}\text{Mpc}$ for $k = 0.1\text{--}0.3h\text{Mpc}^{-1}$ and narrows with k . The local approximation is the limit $K_k \rightarrow \delta_D$.
2. **Edges and holes are handled together.** Near an edge or a hole, $K_k \star m$ is diluted because part of the kernel falls outside the survey. This is the physical loss of pairs straddling the edge, not an estimator bias.

5 The mean and the normalisation

Setting $\mathbf{k}' = \mathbf{k}$ in Equation 5 gives the mean of the estimator:

$$\langle \hat{P}(k) \rangle = G_P(\mathbf{k}, \mathbf{k}) / \text{norm} = P(k) I_k / \text{norm}, \quad I_k = \int d^3x m(\mathbf{x})(K_k \star m)(\mathbf{x}) = \int W_k. \quad (13)$$

The measured power is diluted by the window by the k -dependent factor I_k / norm . The local approximation would give $\int m^2 / \text{norm}$ instead, larger by the same hole and edge effects (for DESI LRG1, $\int m^2 / \text{norm} = 1.22$, while a window resolved and diluted at $7'$ gives 1.03).

6 Corrections for the two choices of model power spectrum

The three terms of Equation 7 are evaluated as

$$\begin{aligned} PP : & P(k)P(k') |\tilde{W}_k(\mathbf{k} - \mathbf{k}')|^2, \\ PS : & P(k) \tilde{W}_k(\mathbf{k} - \mathbf{k}') \tilde{S}^*(\mathbf{k} - \mathbf{k}') + \text{c.c.}, \\ SS : & |\tilde{S}(\mathbf{k} - \mathbf{k}')|^2, \end{aligned} \quad (14)$$

all divided by norm^2 . S is local (self-pairs) and needs no kernel.

6.1 Case 1: theory power spectrum

Input: $P_{\text{th}}(k)$, the true redshift-space galaxy power, unconvolved.

clustering window	$W_k = m(K_k \star m)$ in PP and in the clustering leg of PS
shot-noise window	S unchanged: each random's own w^2 , $\int S = \text{num_shotnoise}$
normalisation	the files' norm (the estimator's), not $\int m^2$ and not I_k
model power	P_{th} as is, with no rescaling. It must be the full power the estimator sees: nonlinear, biased, redshift-space, and any non-Poisson stochasticity, since the files subtract only the realised Poisson shot noise.

6.2 Case 2: measured power spectrum

Input: the mean of the mocks, $\bar{P}(k) \approx P(k) I_k / \text{norm}$ (Equation 13), shot noise subtracted.

clustering window	$W_k = m(K_k \star m)$, as in case 1
shot-noise window	as in case 1
normalisation	the files' norm, as in case 1
model power	$P_{\text{model}}(k) = \bar{P}(k) \text{norm} / I_k$, which recovers $P(k)$ and makes case 2 identical to case 1. In <code>thecov</code> this is automatic: its model correction divides by its own window integral $I = \alpha \sum_r w_r \text{NW}_r$, which equals I_k once $\text{NW} = K_k \star m$.
not undone	the window convolution also changes the <i>shape</i> of $\bar{P}$: smearing of the first bins and of the BAO feature, the integral constraint, and leakage between multipoles. Its effect on the covariance is about twice the relative error in P , small for $k \gtrsim 0.05 h \text{Mpc}^{-1}$. To remove it, deconvolve: fit the P whose window-convolved multipoles equal $\bar{P}_\ell$.

6.3 What thecov did, and why the error was small

`thecov` used $\bar{P} \text{norm} / \int m^2$ as the model and $W = m^2$ as the window. The first makes the model P too low by $I_k / \int m^2 \approx 0.80\text{--}0.85$; the second makes the PP window too large. The two errors have opposite signs and nearly cancel, leaving the $\sim 7\%$ deficit measured below.

7 Evidence from the DESI DR2 validation (LRG1, holi v3, 859 mocks)

Exact Gaussian reference. Gaussian random fields δ with the model power, multiplied by $m(\mathbf{x})$ on a $6h^{-1}\text{Mpc}$ mesh (with m from a healpix map at nside 512 times the weighted $n(z)$, from 10 random files), plus white noise of density S , estimated as in the files, 1000 realisations per cap: this evaluates Equation 7 with no local approximation. With the input power calibrated so that the mean equals the mocks', the calibration factor $\langle \hat{P} \rangle_{\text{mocks}} / \langle \hat{P} \rangle_{\text{GRF}}$ is the measured k -dependence of I_k relative to thecov's $\int m^2$:

	0.02–0.05	0.05–0.1	0.1–0.2	0.2–0.3
calibration NGC	1.276	1.242	1.214	1.202
calibration SGC	1.262	1.220	1.193	1.179
Var GRF / thecov NGC	1.094	1.038	1.064	1.050
Var GRF / thecov SGC	1.071	1.099	1.074	1.059

The variance ratio is independent of the bin width ($\Delta k = 0.005, 0.01, 0.02$), i.e. a Gaussian-term error; the GRF with the high-resolution window and with thecov's smooth mean weight (both diluted by holes) agree to 0.2%, so fine structure of $\langle w \rangle$ plays no role.

Top-hat dilution. Replacing thecov's m at each random by m times the footprint fill fraction of a healpix pixel (a top-hat stand-in for K_k) raises the variance towards the exact result and converges at pixels of $\sim 10h^{-1}\text{Mpc}$:

pixel	none	7'	14'	28'	56'
P0 variance / default, NGC, $k < 0.1$	1	1.044–1.048	1.056–1.061	1.063–1.068	1.068–1.075
P0 variance / default, SGC, $k < 0.1$	1	1.041–1.043	1.058–1.060	1.073–1.076	1.084–1.087
$\langle \chi^2 \rangle / n$ GCcomb, $\Delta k = 0.005$	1.098	1.067	1.051	1.046	1.040

With the 28' pixel and more pair counts (the sampling noise of the pair counts inflates χ^2 by ~ 0.01), $\langle \chi^2 \rangle / n = 1.024$ (NGC), 1.032 (SGC), 1.026 (GCcomb), down from 1.10; the remaining excess is non-Gaussian (it grows with the bin width). The top-hat scale is the arbitrary element that the kernel K_k of Equation 11 removes.

8 Implementation

Only the value of m given to each random (NW) changes; pair counting, windows and covariance assembly are untouched.

1. **Kernel.** From the model monopole $P_0(k)$ (theory, or $\bar{P}_0$ norm/ I iterated once), compute $\xi_0(r)$ by a Hankel transform and $K_k(r) = \xi_0(r)j_0(kr)/P_0(k)$, truncated at $R_{\text{max}} \approx 150h^{-1}\text{Mpc}$ and renormalised to unit integral. Use the band-averaged kernel for each of $N_b = 3\text{--}4$ k -bands spanning $0.02\text{--}0.3h\text{Mpc}^{-1}$.
2. **Smoothed density.** Paint $\alpha \sum_r w_r \delta_D(\mathbf{x} - \mathbf{x}_r)$ from many random files (e.g. 10) on a mesh with cells of $3\text{--}5h^{-1}\text{Mpc}$; convolve with K_k by FFT; interpolate ($K_k \star m$) at thecov's randoms. Painting is safe here: the smoothing is linear and much wider than a cell, so the painting noise averages out (nothing noisy is squared). The painted randoms must be independent of thecov's randoms, so that no random averages with itself (the self-pair would bring back the $\langle w^2 \rangle$ bias).
3. **Windows.** For each band, set $\text{NW}_r = (K_k \star m)(\mathbf{x}_r)$ and compute the window pair counts as now. The S window is unchanged. The window integral is then I_k and the model correction of case 2 is automatic; for case 1, switch it off and pass P_{th} .

4. **Covariance.** Assemble each band's rows and columns from its own windows and interpolate between bands in (k, k') ; blocks with k and k' in different bands use the band of the geometric mean.
5. **Pair-count sampling.** Use enough randoms and near pairs (10^{10} , with the near sample not capped by the randoms kept) for the off-diagonal structure: its noise biases χ^2 high.

Numerical parameters and their convergence tests. Mesh cell (halve it), $R_{\max}$ (increase it), number of k -bands (double it), random files for the painted density (double them), near pairs (double them). None of them is a physical choice.

9 Validation plan and caveats

1. I_k from the kernel must reproduce the GRF calibration curve above, with no mocks involved.
2. `thecov` with $NW = K_k \star m$ must match the GRF variance at every k and bin width, for LRG1 NGC and SGC, and for a second tracer with a different footprint and redshift (ELG1 or BGS).
3. Against the mocks, the bin-width-independent excess should vanish; the remainder is the non-Gaussian covariance (trispectrum, super-sample).

Caveats. (i) The kernel uses the isotropic ξ_0 and j_0 ; the redshift-space ξ and the direction of $\mathbf{k}$ make it anisotropic for the multipole covariances. The top-hat test reduced the P_2 and P_4 excess as much as P_0 's, which suggests the isotropic kernel suffices; the GRF with an anisotropic P can test it. (ii) $k \approx k'$ within a window width, and P smooth across it, as in the original approximation: this remains the low- k limitation. (iii) Case 1 requires the stochastic part of the theory power to match the data's; case 2 inherits it from the mocks.